



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

DATA WAREHOUSING AND MINING [CSE-4060]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

1) Summarize the importance of Data Mining in knowledge Discovery.

(2)

A)

B)

Object	A1	A2
y1	3.0	4.0
y2	2.5	3.5
y3	4.0	2.0
y4	3.5	4.5
y5	2.0	4.0

Using the above data of attributes and objects and given the query point $q = (3.0, 3.0)$:

(a) Calculate the Euclidean distance between the query point and each data point.

(5)

(b) Calculate the Manhattan distance between the query point and each data point.

(c) Determine which point is closest to the query using:

- Euclidean distance
- Manhattan distance

(d) Compute the Cosine similarity between the query point and:

- y1
- y3

C) Suppose a group of 12 sales price records has been sorted as follows:
5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215.

Partition them into three bins by each of the following methods:

(a) equal-frequency (equal-depth) partitioning

(3)

(b) equal-width partitioning

(c) clustering

2) Discuss any two Greedy methods for Attribute Subset selection.

(2)

- A)
- B) Compare and contrast OLTP and OLAP.
Justify the given statement "Data Warehouse is Integrated" with suitable example. (5)
- C) Suppose that a data warehouse consists of the four dimensions date, spectator, location, and game, and the two measures count and charge, where charge is the fare that a spectator pays when watching a game on a given date. Spectators may be students, adults, or seniors, with each category having its own charge rate. (3)

- a) Draw a star schema diagram for the data warehouse.
b) Compare and Contrast of Star schema or a Snowflake schema

- 3) Explain the working principles of Apriori algorithm for finding frequent item sets using suitable example data set. (5)

- A)
- B) Differentiate between support and confidence in Association rule mining.
List and explain three methods of Improving Apriori algorithm. (5)

- 4) **Apply ID3 algorithm and construct Decision tree for the below given data set:**

A)

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Weak	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cold	Normal	Weak	Yes
D10	Rain	Mild	Normal	Strong	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

(5)

Classify for new instance (Outlook = Sunny, Humidity = Normal, Wind = Strong).

- B) Determine the working principles involved in support vector machine algorithm for any classification task. (5)
- 5) Explain the working steps of algorithm for k-Medoids: A Representative Object-Based Technique with its advantages and limitations. (5)

A)

- B) List the steps of DBSCAN algorithm and discuss how the parameters eps and minPts affect the performance of the DBSCAN algorithm. Include visual illustrations where applicable. (5)